

EXAM #3Name Key
(print)

Total = 100 points

Please show all your work.

1. (20 pts) Fifty-five percent of adults say cashews are their favorite kind of nut. You randomly select 12 adults and ask each to name his or her favorite nut. Find the probability that the number who say cashews are their favorite nut is:

a) exactly 7 ${}_{12}C_7 \cdot .55^7 \cdot .45^5 = .0222$

Answer 22.2%

b) at most 2

$${}_{12}C_2 \cdot .55^2 \cdot .45^{10} + {}_{12}C_1 \cdot .55 \cdot .45^{11} + {}_{12}C_0 \cdot .55^0 \cdot .45^{12}$$

$$.0070 + 0.0010 + 0.0001 = .0081$$

Answer .81%

2. (20 pts) A study found that the mean height of adult males is 69 inches with a standard deviation of 3 inches. Assume that the heights are normally distributed.

a) (10 pts) Find the probability that the height of a randomly selected adult male is between 64 and 70 inches?

$$z_{64} = \frac{64 - 69}{3} = -1.67$$

$$z_{70} = \frac{70 - 69}{3} = .33$$

$$P(64 \leq x \leq 70) = P(x \leq 70) - P(x \leq 64)$$

$$= P(z \leq .33) - P(z \leq -1.67) =$$

$$= .6293 - .0475 = .5818$$

Answer 58.18%

b) (10 pts) If you randomly select 400 adult males, about how many of them will be over 73 inches tall?

$$z_{73} = \frac{73 - 69}{3} = 1.33$$

$$P(x > 73) = P(z > 1.33) = 1 - P(z \leq 1.33)$$

$$= 1 - .9082 = 0.0918$$

$$0.0918 \times 400 = 36.72$$

Answer 37

3. (25 pts) One in four adults say he/she has no trouble sleeping at night. You randomly select five adults and ask if he/she has no trouble sleeping at night.

a)(10 pts) Construct the probability distribution (for this sample of five adults, where x represents the number of adults that have no trouble sleeping at night).

x	$P(x)$
0	.237
1	.396
2	.264
3	.088
4	.015
5	.001

$$P(0) = .75^5$$

$$P(1) = 1C5 \cdot .25 \times .75^4$$

$$P(2) = 2C5 \cdot .25^2 \times .75^3$$

$$P(3) = 3C5 \cdot .25^3 \times .75^2$$

$$P(4) = 4C5 \cdot .25^4 \times .75$$

$$P(5) = .25^5$$

b) (7 pts) Find the probability that the number of people (from this group of five randomly selected adults) who say that they have no trouble sleeping is exactly 2.

Answer 26.4 %

c) (8 pts) Find the mean and the standard deviation.

$$\mu = .25 \times 5 = 1.25$$

$$\sigma = \sqrt{5 \times .25 \times .75} = .97$$

Mean 1.25

Standard deviation .97

4. (25 pts) The average number of cups of coffee per day that each student drinks is given below. Construct a probability distribution and then find the mean and the standard deviation of the probability distribution.

Cups	Students	$P(x)$	$(x-M)$	$(x-M)^2$
0	21	.266	-1.635	2.673
1	16	.203	-.635	.403
2	20	.253	.365	.133
3	15	.190	1.365	1.863
4	7	.089	2.365	5.593
	<u>79</u>			

$$M = \sum x P(x) = 1.635$$

$$G = \sqrt{\sum (x-M)^2 P(x)} = 1.295$$

Mean 1.6

Standard deviation 1.3

5. (10 pts) 1,500 raffle tickets are sold at \$2 each for 4 prizes valued at \$800, \$600, \$500, and \$300. You buy one ticket. What is the expected value of your gain/loss?

$$M = 798 \times \frac{1}{1500} + 598 \times \frac{1}{1500} + 498 \times \frac{1}{1500} + 298 \times \frac{1}{1500} + -2 \times \frac{1496}{1500}$$

$$= -0.533$$

Answer -.53 \$

Name Key
(print)

Total = 100 points

Please show all your work.

1. (20 pts) Forty-five percent of adults say cashews are their favorite kind of nut. You randomly select 12 adults and ask each to name his or her favorite nut. Find the probability that the number who say cashews are their favorite nut is:

a) exactly 7 $12C_7 \cdot .45^7 \times .55^5 = 0.149$

Answer 14.9%

b) at most 2

$$12C_2 \cdot .45^2 \times .55^{10} + 12C_1 \cdot .45 \times .55^{11} + 12C_0 \times .55^{12}$$

$$.0339 + 0.0075 + 0.0008 = 0.0422$$

Answer 4.22%

2. (20 pts) A study found that the mean height of adult females is 66 inches with a standard deviation of 2 inches. Assume that the heights are normally distributed.

a)(10 pts) Find the probability that the height of a randomly selected adult female is between 64 and 70 inches?

$$z_{64} = \frac{64-66}{2} = -1 \quad P(64 \leq x \leq 70) = P(-1 \leq z \leq 2) = P(z \leq 2) - P(z \leq -1)$$

$$= .9772 - .1587 = .8185$$

$$z_{70} = \frac{70-66}{2} = 2$$

Answer 81.85%

b)(10 pts) If you randomly select 400 adult females, about how many of them will be over 70 inches tall?

$$z_{70} = 2 \quad P(x \geq 70) = P(z \geq 2) = 1 - .9772 = 0.0228$$

$$400 \times 0.0228 = 9.12 \approx 9$$

Answer 9

3. (25 pts) One in three adults say he/she has no trouble sleeping at night. You randomly select five adults and ask if he/she has no trouble sleeping at night.

a)(10 pts) Construct the probability distribution (for this sample of five adults, where x represents the number of adults that have no trouble sleeping at night).

x	$P(x)$
0	.135
1	.332
2	.328
3	.161
4	.040
5	.004

$$P(0) = .67^5$$

$$P(1) = 5C_1 \cdot .33 \times .67^4$$

$$P(2) = 5C_2 \cdot .33^2 \times .67^3$$

$$P(3) = 5C_3 \cdot .33^3 \times .67^2$$

$$P(4) = 5C_4 \cdot .33^4 \times .67$$

$$P(5) = .33^5$$

b) (7 pts) Find the probability that the number of people (from this group of five randomly selected adults) who say that they have no trouble sleeping is exactly 2.

Answer .328 = 32.8%

c) (8 pts) Find the mean and the standard deviation.

$$M = .33 \times 5 = 1.67$$

$$G = \sqrt{5 \times .33 \times .67} = 1.05$$

Mean 1.67

Standard deviation 1.05

4. (25 pts) The average number of cups of coffee per day that each student drinks is given below. Construct a probability distribution and then find the mean and the standard deviation of the probability distribution.

Cups	Students	$P(x)$	$(x - \mu)$	$(x - \mu)^2$
0	17	.215	-1.714	2.935
1	18	.228	-.714	.517
2	22	.278	.281	.079
3	14	.177	1.281	1.641
4	8	.101	2.281	5.203
	<u>79</u>			

$$\mu = \sum x P(x) = 1.714$$

$$\sigma = \sqrt{\sum (x - \mu)^2 P(x)} = 1.26$$

Mean 1.714

Standard deviation 1.26

5. (10 pts) 1,000 raffle tickets are sold at \$2 each for 4 prizes valued at \$800, \$600, \$500, and \$300. You buy one ticket. What is the expected value of your gain/loss?

$$798 \times \frac{1}{1000} + 598 \times \frac{1}{1000} + 498 \times \frac{1}{1000} + 298 \times \frac{1}{1000} + (-2) \times \frac{996}{1000} = -.20$$

Answer -.20 \$